

Thermodynamics

Lecture 4A:

- Energy analysis of closed systems

Prof. Claudio Hail (he/him)

Recap Lecture 3B

Table interpolations

$$v(T) = v_1 + (T - T_1) \frac{(v_2 - v_1)}{(T_2 - T_1)}$$

Compressed liquid

$$v \approx v_f(T)$$

$$u \approx u_f(T)$$

Ideal gas law

$$pV = n\bar{R}T \quad \bar{R} = 8.314 \frac{J}{molK}$$

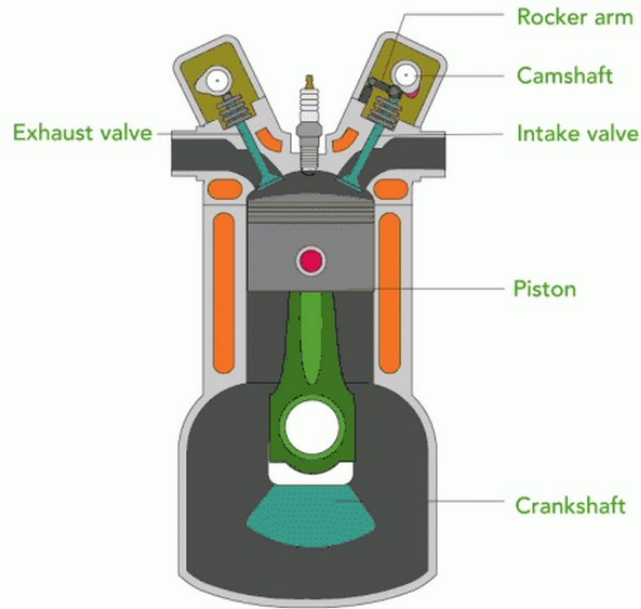
$$pV = mRT$$

- Valid at low pressure high temperature
- Negligible intermolecular forces

After this lecture

- Understand how to describe different cases of work due to moving boundaries.
- Perform a general energy analysis on closed systems.

Motivation



Moving boundary work (I)

Differential work due to a boundary displacement

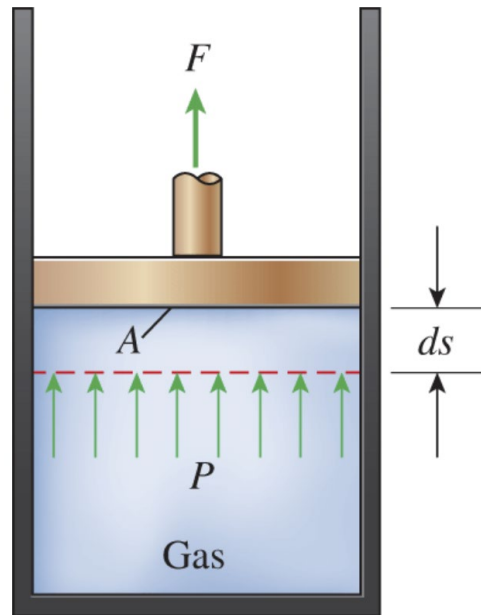
$$\delta W_b = F \delta s$$
$$dV > 0 \text{ Expansion}$$
$$dV < 0 \text{ Compression}$$

Quasi-static process:

$$\delta W_b = F ds = p A ds = p dV$$

Total work:

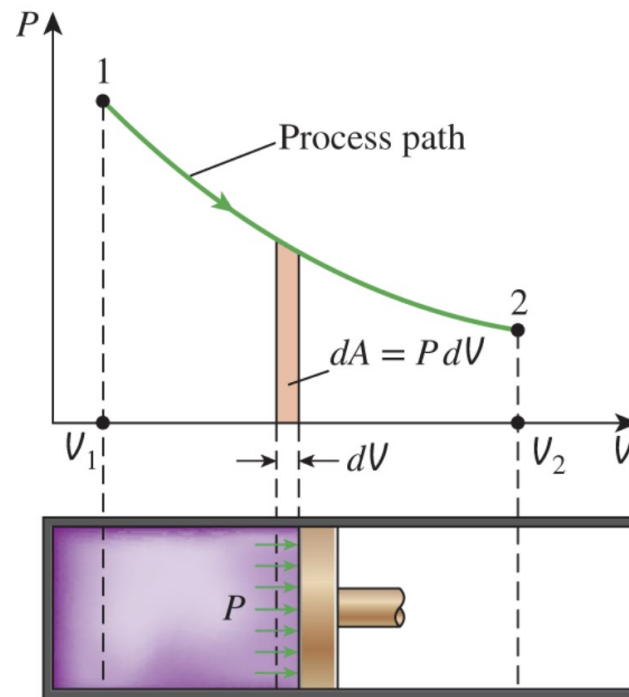
$$W_b = \int_1^2 p dV$$



Moving boundary work (II)

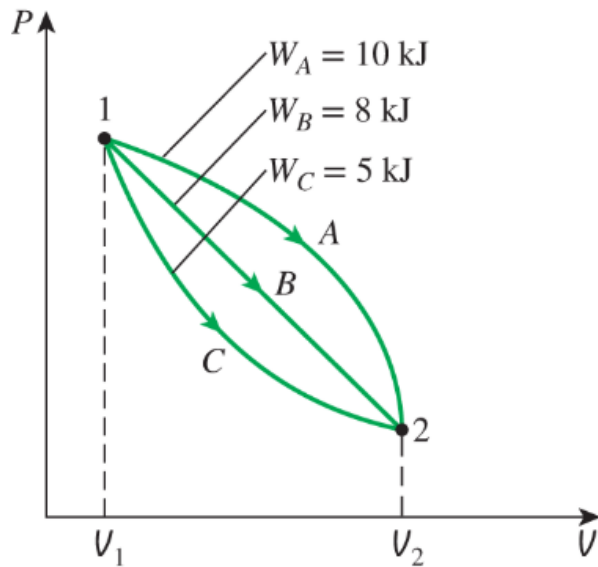
- In a P-V diagram the area under the curve represents the work transfer during the process.

$$A = \int_1^2 dA = \int_1^2 p dV$$

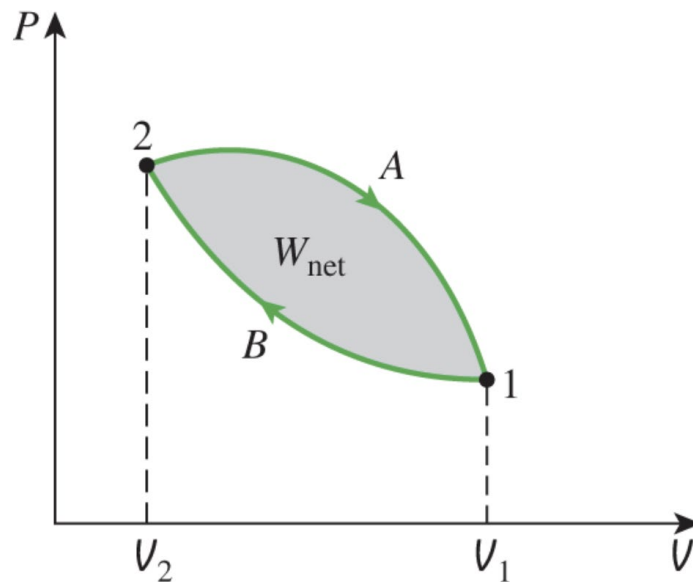


Moving boundary net work

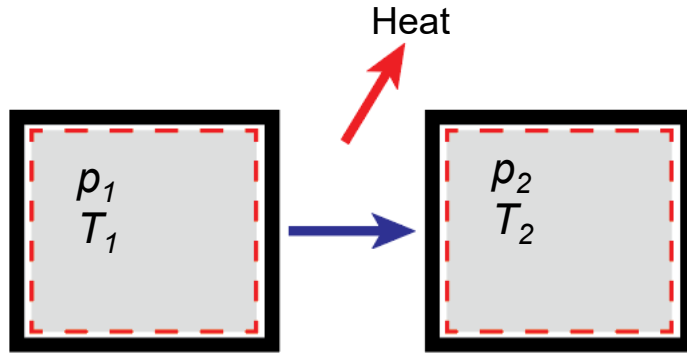
- Comparison of different processes



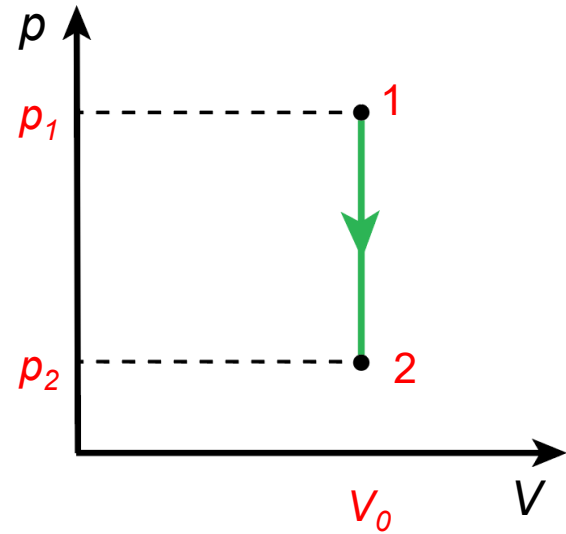
- Net work of a cycle



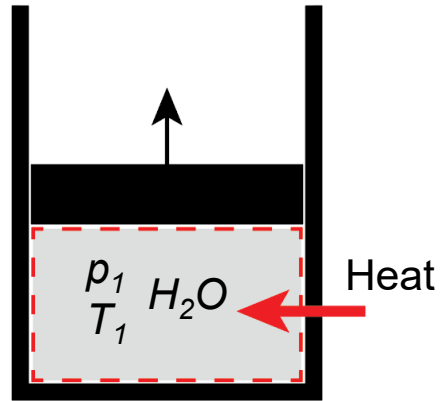
Work: Constant-volume process



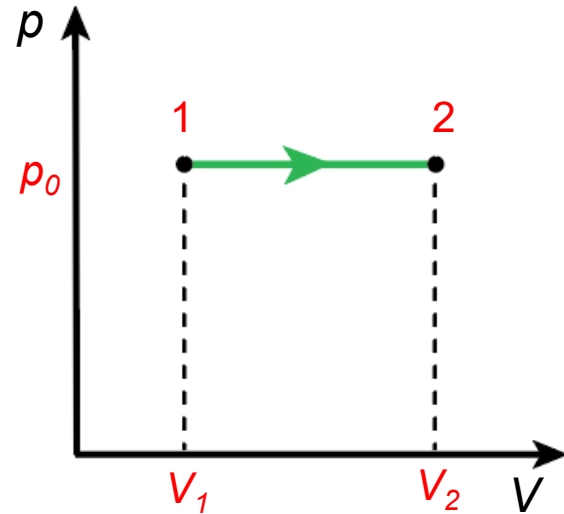
$$W_b = \int_{V_1}^{V_2} p dV = 0$$



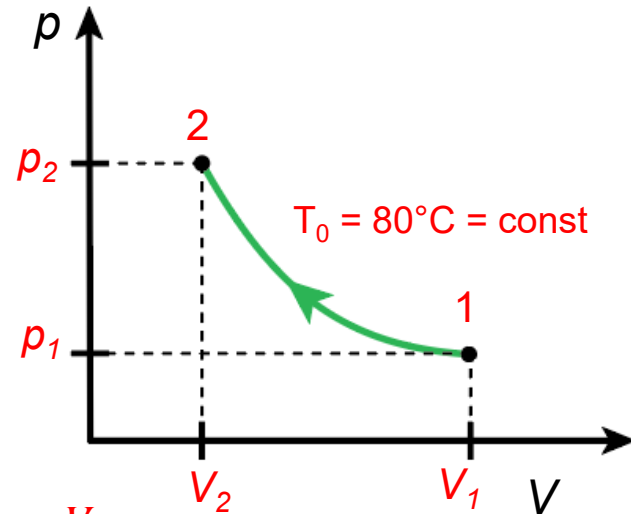
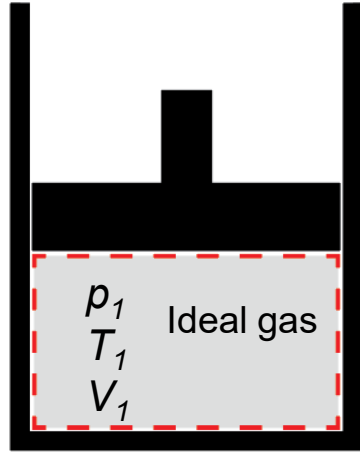
Work: Constant-pressure process



$$W_b = \int_{V_1}^{V_2} p dV = p_0 \int_{V_1}^{V_2} dV = p_0(V_2 - V_1)$$



Work: Constant-temperature process for ideal gas



$$p_1 V_1 = mRT_1 = \text{const.}$$

$$W_b = \int_{V_1}^{V_2} p dV = \int_{V_1}^{V_2} \frac{mRT_1}{V} dV = mRT_1 \ln \frac{V_2}{V_1}$$

Polytropic processes

- A general thermodynamic process following the relation

$$pV^n = \text{const.}$$

Polytropic index n

Isobaric

$$n = 0$$

Isothermal

$$n = 1$$

Adiabatic/isentropic

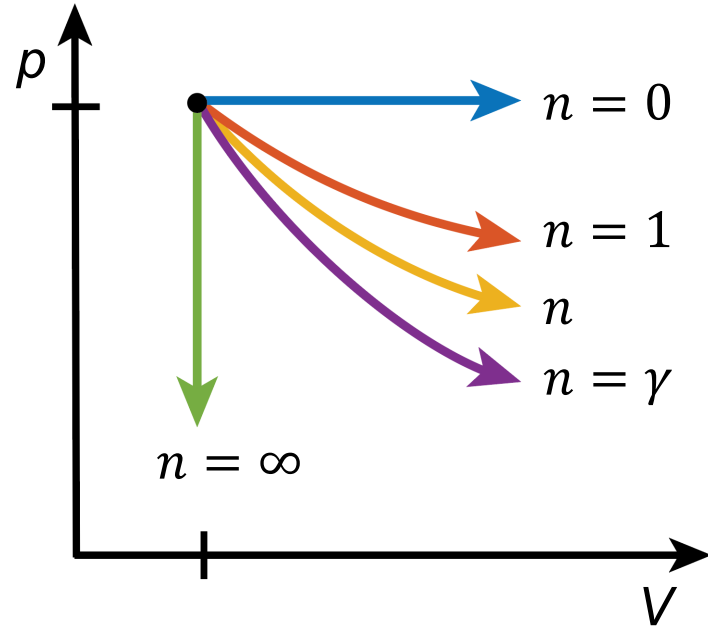
$$n = \gamma = c_p/c_v$$

Isochoric

$$n = \infty$$

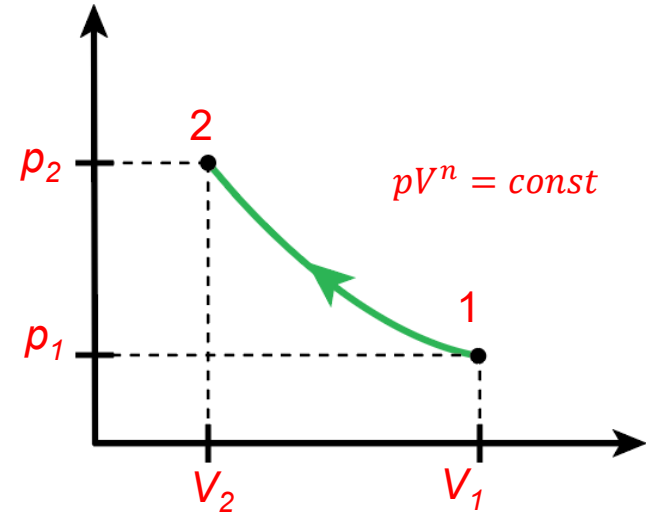
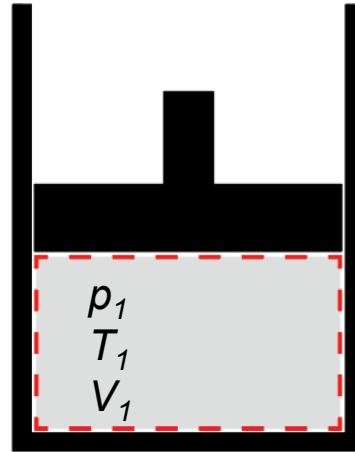
Polytropic

$$n$$



Work: Polytropic process

$$p_1 V_1^n = p_2 V_2^n = \text{const.}$$



$$W_b = \int_{V_1}^{V_2} \frac{p_1 V_1^n}{V^n} dV = p_1 V_2 \frac{V_2^{1-n} - V_1^{1-n}}{-n + 1} = \frac{p_2 V_2 - p_1 V_1}{1 - n}$$

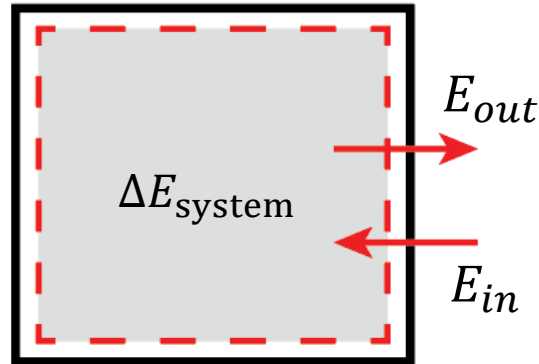
Energy balance for closed systems

Closed system

$$E_{in} - E_{out} = \Delta E_{\text{system}}$$

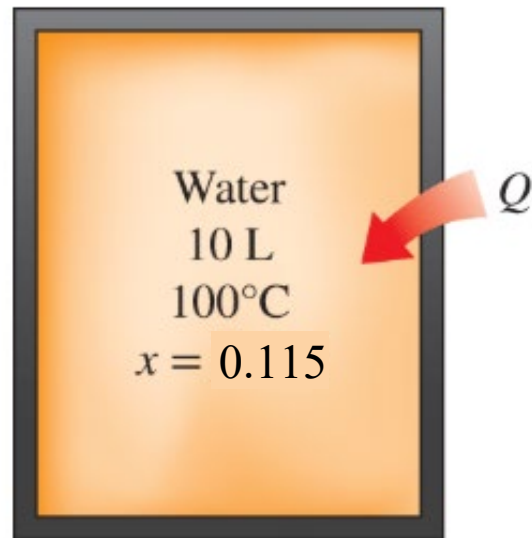
Closed stationary system

$$\Delta U = Q - W$$



Example I

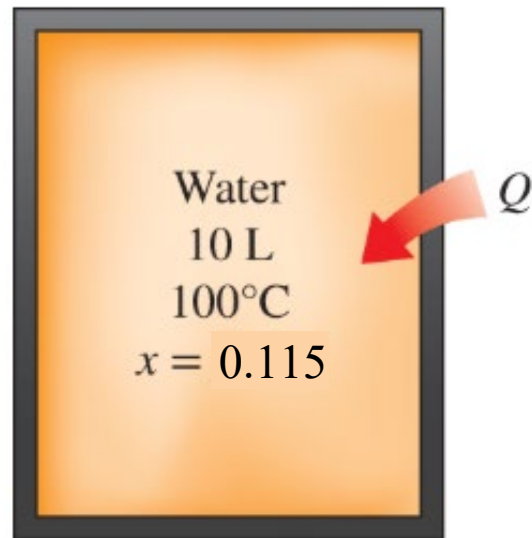
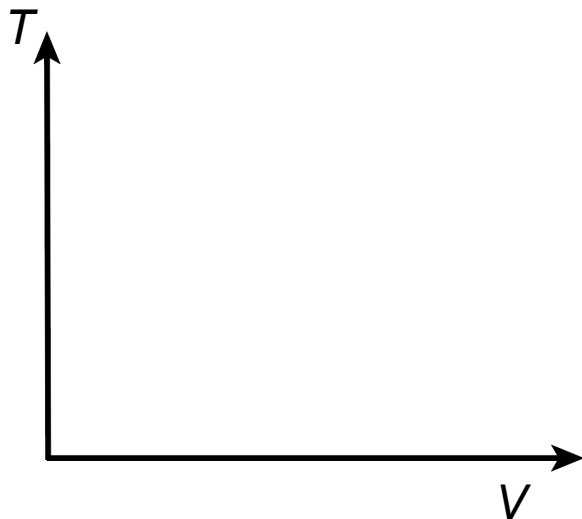
- A rigid 10-L vessel initially contains a mixture of liquid water and vapor at 100°C with 11.5% quality. The mixture is then heated until its temperature is 180°C. Calculate the heat transfer required for this process



Example I

$$T_1 = 100^\circ\text{C} ; V_1 = 0.01\text{m}^3 ; x_1 = 0.115$$

$$T_2 = 180^\circ\text{C} ; V_2 = ? ;$$



Example I

$$T_1 = 100^\circ\text{C} ; V_1 = 0.01\text{m}^3 ; x_1 = 0.115$$

$$T_2 = 180^\circ\text{C} ; V_2 = ? ;$$

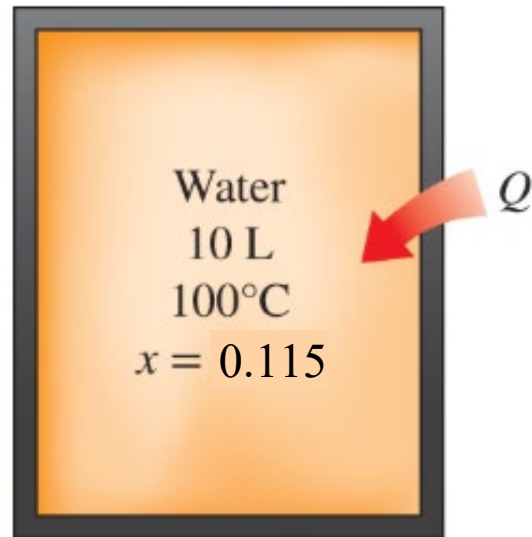
Energy Analysis:

$$E_{in} - E_{out} = \Delta E_{\text{system}}$$

$$Q - W = \Delta U \quad m = \frac{V_1}{v_1} = 0.0517 \text{ kg}$$

$$Q = U_2 - U_1 = m(u_2 - u_1)$$

$$Q = 99.25 \text{ kJ}$$



Example I

State 1

$$T_1 = 100^\circ\text{C} ; V_1 = 0.01\text{m}^3 ; x_1 = 0.115$$

$$u_1 = (1 - x)u_f + xu_g = 659.058 \text{ kJ/kg}$$

$$v_1 = (1 - x)v_f + xv_g = 0.1932 \text{ kg/m}^3$$

Saturated water—Temperature table

Temp., T °C	Sat. Press., P_{sat} kPa	Specific volume, m^3/kg		Internal energy, kJ/kg		
		Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g
85	57.868	0.001032	2.8261	355.96	2131.9	2487.8
90	70.183	0.001036	2.3593	376.97	2117.0	2494.0
95	84.609	0.001040	1.9808	398.00	2102.0	2500.1
100	101.42	0.001043	1.6720	419.06	2087.0	2506.0
105	120.90	0.001047	1.4186	440.15	2071.8	2511.9
110	143.38	0.001052	1.2094	461.27	2056.4	2517.7
115	169.18	0.001056	1.0360	482.42	2040.9	2523.3

Example I

State 2

$$T_2 = 180^\circ\text{C} ; V_2 = ? ; x_2 = ?$$

$$v_2 = v_1 = 0.1932 \frac{\text{kg}}{\text{m}^3} > v_g$$

$$x_2 = \frac{v_2 - v_f}{v_g - v_f} = 0.99667$$

$$u_2 = 2576.736 \text{ kJ/kg}$$

Saturated water—Temperature table

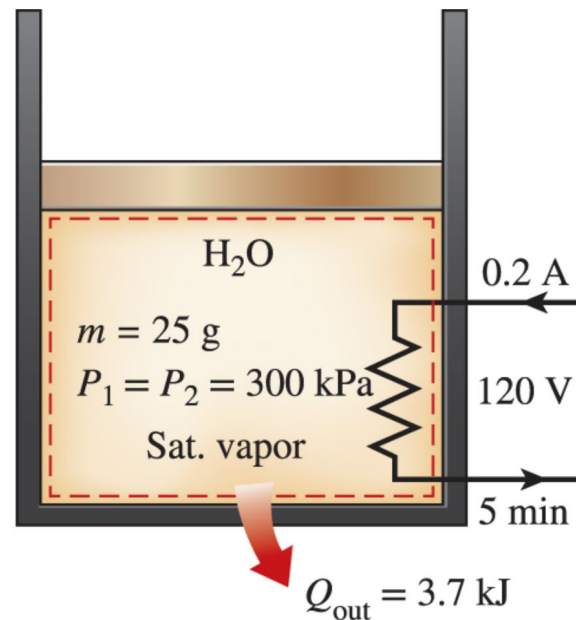
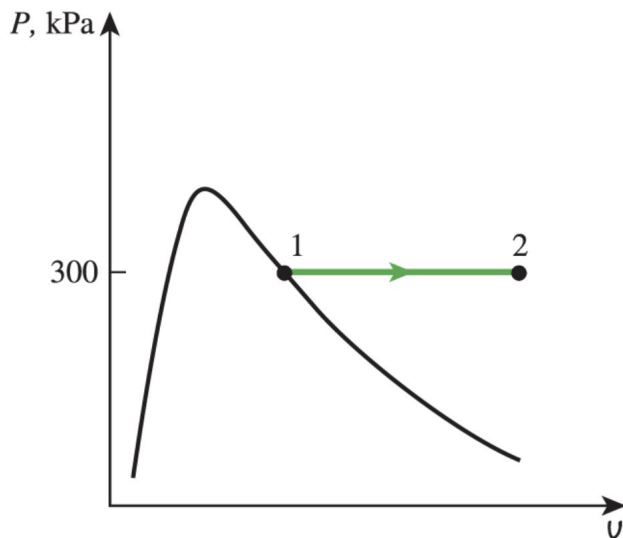
Temp., T °C	Sat. Press., P_{sat} kPa	Specific volume, m^3/kg		Internal energy, kJ/kg		
		Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g
170	792.18	0.001114	0.24260	718.20	1857.5	2575.7
175	892.60	0.001121	0.21659	740.02	1839.4	2579.4
180	1002.8	0.001127	0.19384	761.92	1820.9	2582.8
185	1123.5	0.001134	0.17390	783.91	1802.1	2586.0
190	1255.2	0.001141	0.15636	806.00	1783.0	2589.0
195	1398.8	0.001149	0.14089	828.18	1763.6	2591.7

Summary

- Understand how to describe different cases of work due to moving boundaries.
- Perform a general energy analysis on closed systems.

Example II

- Constant pressure electrical heating of water



Example II

- Constant pressure electrical heating of water

$$E_{in} - E_{out} = \Delta E_{system}$$

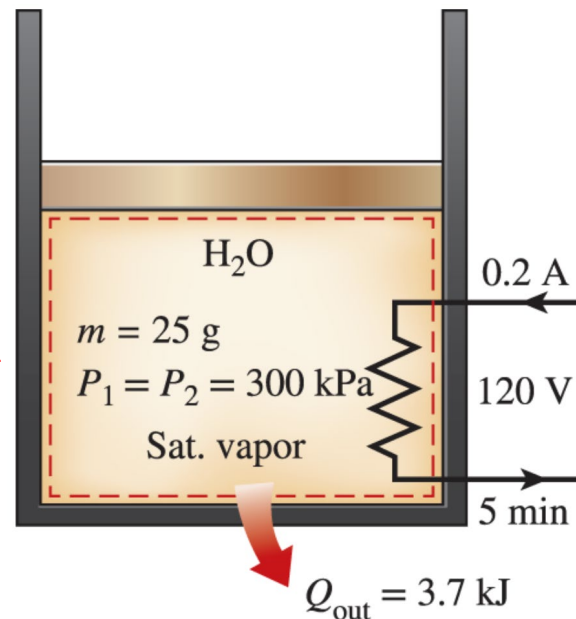
$$Q - W = \Delta U$$

$$Q - W_{other} - W_b = U_2 - U_1$$

$$Q - W_{other} - (p_2V_2 - p_1V_1) = U_2 - U_1$$

$$Q - W_{other} = H_2 - H_1 = m(h_2 - h_1)$$

$$W_{other} = VI\Delta t = 7200 \text{ J}$$



Example II

- State 1:
 - $p_1 = 300 \text{ kPa}$
 - $x = 1$

Table A-5

Press., P kPa	Sat. temp., T_{sat} $^{\circ}\text{C}$	Specific volume, m^3/kg		Internal energy, kJ/kg			Enthalpy, kJ/kg		
		Sat. liquid, u_f	Sat. vapor, u_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g
200	120.21	0.001061	0.88578	504.50	2024.6	2529.1	504.71	2201.6	2706.3
225	123.97	0.001064	0.79329	520.47	2012.7	2533.2	520.71	2191.0	2711.7
250	127.41	0.001067	0.71873	535.08	2001.8	2536.8	535.35	2181.2	2716.5
275	130.58	0.001070	0.65732	548.57	1991.6	2540.1	548.86	2172.0	2720.9
300	133.52	0.001073	0.60582	561.11	1982.1	2543.2	561.43	2163.5	2724.9
325	136.27	0.001076	0.56199	572.84	1973.1	2545.9	573.19	2155.4	2728.6
350	138.86	0.001079	0.52422	583.89	1964.6	2548.5	584.26	2147.7	2732.0
375	141.30	0.001081	0.49133	594.32	1956.6	2550.9	594.73	2140.4	2735.1

Example II

- State 2:
 - $p_2 = 300 \text{ kPa}$
 - $h_2 = 2864.9 \text{ kJ/kg}$

$$T_2 \approx 200^\circ\text{C}$$

$$T_2 = 199.52^\circ\text{C}$$

Table A-6

T $^\circ\text{C}$	v m^3/kg	u kJ/kg	h kJ/kg	s $\text{kJ/kg}\cdot\text{K}$	v m^3/kg	u kJ/kg	h kJ/kg	s $\text{kJ/kg}\cdot\text{K}$
$P = 0.20 \text{ MPa} (120.21^\circ\text{C})$					$P = 0.30 \text{ MPa} (133.52^\circ\text{C})$			
Sat.	0.88578	2529.1	2706.3	7.1270	0.60582	2543.2	2724.9	6.9917
150	0.95986	2577.1	2769.1	7.2810	0.63402	2571.0	2761.2	7.0792
200	1.08049	2654.6	2870.7	7.5081	0.71643	2651.0	2865.9	7.3132
250	1.19890	2731.4	2971.2	7.7100	0.79645	2728.9	2967.9	7.5180
300	1.31623	2808.8	3072.1	7.8941	0.87535	2807.0	3069.6	7.7037
400	1.54934	2967.2	3277.0	8.2236	1.03155	2966.0	3275.5	8.0347
500	1.78142	3131.4	3487.7	8.5153	1.18672	3130.6	3486.6	8.3271
600	2.01302	3302.2	3704.8	8.7793	1.34139	3301.6	3704.0	8.5915
700	2.24434	3479.9	3928.8	9.0221	1.49580	3479.5	3928.2	8.8345

Next up

- Internal energy, enthalpy, and specific heats of ideal gases.
- Internal energy, enthalpy, and specific heats of liquids and solids.

Learning objectives

What you will be able to do after this course:

- I. *Knowledge:* Write the first and second laws of thermodynamics.**
- II. *Comprehension:* Describe the first and second laws in the student's own language.**
- III. *Application:* Solve simple single-answer problems using the first law.**
- IV. *Analysis:* Solve problems requiring both the first and second laws.
- V. *Synthesis:* Develop schemes to improve the efficiency or performance of thermodynamic systems, e.g., compressors, turbines, heat pumps, etc.

Learning objectives (II)

- VI. Evaluation: Determine and describe second law fallacies in proposed power cycles.
- VII. Evaluation: Judge when classical thermodynamics is not the appropriate analysis tool.
- VIII. Evaluation: Find and correct errors in the student's own solutions and in those of others.
- IX. *Analysis and Evaluation:* Search appropriate databases and the literature to find required thermodynamic data, and if the data are not available the student can select appropriate procedures and predict the values of the data.**
- X. Knowledge: Briefly outline the history of the field of thermodynamics.